

LCSNet: Lightweight Caries Segmentation Network for the segmentation of dental caries using smartphone photographs

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ABSTRACT

Dental caries is one of the major dental issues that is common among many individuals. It leads to tooth loss and affects the tooth root, creating a need to automatically detect dental caries to reduce treatment costs and prevent its consequences. The Lightweight Caries Segmentation Network (LCSNet) proposed in this study detects the location of dental caries by applying pixel-wise segmentation to dental photographs taken with various Android phones. LCSNet utilizes a Dual Multiscale Residual (DMR) block in both the encoder and decoder, adapts transfer learning through a pre-trained InceptionV3 model at the bottleneck layer, and incorporates a Squeeze and Excitation block in the skip connection, effectively extracting spatial information even from images where 95 % of the background and only 5 % represent the area of interest. A new dataset was developed by gathering oral photographs of dental caries from two hospitals, with advanced augmentation techniques applied. The LCSNet architecture demonstrated an accuracy of 97.36 %, precision of 73.1 %, recall of 70.2 %, an F1-Score of 71.14 %, and an Intersection-over-Union (IoU) of 56.8 %. Expert dentists confirmed that the LCSNet model proposed in this in vivo study accurately segments the position and texture of dental caries. Both qualitative and quantitative performance analyses, along with comparative analyses of efficiency and computational requirements, were conducted with other deep learning models. The proposed model outperforms existing deep learning models and shows significant potential for integration into a smartphone application-based oral disease detection system, potentially replacing some conventional clinically adapted methods.

1. Introduction

Oral health problems are widespread and increasingly common, leading to pain and discomfort that can affect daily life [1]. According to the WHO Global Oral Health Status Report (2022), nearly 3.5 billion people worldwide suffer from oral diseases [2]. Among these, dental caries is one of the most prevalent infectious conditions, which can cause infection, pain, and eventual tooth loss if left untreated [3–5]. Early detection of dental caries is vital because many individuals tend to delay seeking treatment until symptoms become severe, making prompt diagnosis essential for effective and affordable care [6].

Currently, clinicians primarily rely on visual examinations to identify early signs of caries, such as changes in enamel and structural breakdown, often supplemented with radiographs for confirmation [7].

These conventional methods require skilled professionals, and without their expertise, there is a higher risk of misdiagnoses. In underdeveloped and developing regions, primary health centers frequently lack dental specialists, compelling patients to travel long distances to access dental care, which imposes a significant financial strain on rural populations. This situation underscores the need for more accessible and cost-effective screening tools for early caries detection.

Advancements in artificial intelligence (AI) are increasingly transforming dental diagnostics by aiding in diagnosis, treatment planning, and patient management [60,61]. Many smartphones now feature affordable, high-quality cameras, enabling rural communities to capture images of the oral cavity and transmit them to dental practitioners at district hospitals for remote consultation. This approach could decrease travel costs and improve access to dental care [17]. To address these

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challenges, we propose a deep learning-based segmentation model tailored for analyzing images captured via smartphones. Automating the segmentation process enables a detailed assessment of dental conditions, highlighting areas of decay and severity crucial for accurate caries classification [6,12]. Segmenting dental images is helpful for thoroughly analyzing dental conditions [8–11]; it provides more information about the decayed areas and the degree of decay, assisting in further caries classification [6,12]. UNET [13] is the first DL architecture designed for medical image segmentation using encoder-decoder blocks and is capable of extracting complex features through a fusion layer [14].

This study aims to develop a lightweight encoder-decoder architecture for automatically segmenting dental caries using photographs taken with smartphone cameras and to evaluate its performance against other existing models. The proposed model utilizes fewer layers and incorporates DMR blocks in both the encoder and decoder, adapting a transfer-learning approach with the InceptionV3 pre-trained model in the bottleneck layer and incorporating SE blocks in the skip connections. Compared to existing models such as UNET [13], UNET-VGG16 [53], UNET-Resnet50 [49], UNET-InceptionV3 [50], UNET-InceptionResnetv2 [51], TransAttUnet [66], ResnetXt [67], and SelfAttention [68], our proposed LCSNet has fewer trainable parameters, making it suitable for deployment on devices with limited resources. Performance evaluation against these models demonstrates its effectiveness in accurately segmenting dental caries from smartphone images. The key contributions of this work are:

- Images from the hospital are collected according to the proper study protocol to prepare a new dataset for use in this study.
- The proposed LCSNet model deploys three encoders and three decoders, each containing DMR blocks. The pre-trained InceptionV3 is utilized in the bottleneck layer and incorporates a squeeze-and-excitation block in the skip connection.
- The proposed lightweight model is trained with only 5.87 million parameters, reducing the architectural complexity.
- Experiments were conducted with other deep learning segmentation models, and the resulting performance metrics were compared. The outcomes of the proposed and other deep learning models were analyzed quantitatively and qualitatively.

Following the introduction, Section 2 provides a comprehensive review of the literature on detecting dental caries. Section 3 describes the adapted materials and methods and the performance of the models. In Section 4, a detailed comparison is made between the results of the developed model and other segmentation models. Finally, it concludes that the predictive results of the developed LCSNet model are accurate and could be utilized in clinical practice.

2. Related work

Many researchers have proposed deep-learning models for detecting dental caries using different diagnostic aids.

2.1. Panoramic dental radiographs

Panoramic dental radiographs provide a wide view of both jaws, facilitating accurate diagnosis. Several deep learning models have been developed to analyze these images. A UNET-based network [15] with axial attention accurately segments caries of varying depths. Another UNET variant [9], enhanced with attention and self-attentive blocks, performs binary segmentation by capturing broader features. A model combining genetic algorithms and capsule networks [16] classifies caries and non-caries teeth with 86.05 % accuracy. All three models are explicitly trained on panoramic images, ensuring tailored performance for this imaging modality. Additionally, a YOLOv8-based system [59] detects multiple dental issues, such as broken roots and missing teeth, with precise localization, even when trained on limited data.

Demonstrates strong diagnostic performance, highlighting its efficiency and robustness.

2.2. Bitewing radiographs

Bitewing radiographs help detect dental decay in the crown areas of both upper and lower teeth. Various deep learning models have shown strong performance in caries detection using these images. A U-shaped deep CNN [18] achieved an F1-score of 64.14 %, while a LeNet model [19] performed better with 91.37 % sensitivity, 98.92 % specificity, and a very low error rate. Advanced segmentation models, such as ResNeXt-RNN [20], achieved 93.67 % accuracy for molars and premolars, while UNET [21] attained a specificity of 0.83. YOLO-based models [22,23] also proved effective, with one reaching 94.59 % [22] accuracy and another reporting a mean average precision of 0.647 for proximal caries [23]. MobileNetV2 [24] classified third molar caries with 87 % accuracy, and AlexNet [25] achieved 90.30 % accuracy by enhancing individual teeth in the radiographs.

2.3. Cone-beam computed tomography(CBCT) images

Multi-input convolutional neural networks [26], along with the combination networks AlexNet and GoogleNet [27], classify caries and non-caries with an accuracy of 95.3 % [26] and 87.4 % [27], respectively. However, acquiring CBCT images from patients undergoing root canal treatment is prohibited, and these images are primarily used to detect vertical root fractures [28,29].

2.4. Near-infrared trans-illumination (NITI) images

Near-infrared trans-illumination (NITI) images are analyzed for the presence of caries using CNN architecture [30,31]. Additionally, the presented NITI imaging is an effective modality for detecting early-stage caries lesions. However, due to the harmful effects of ionizing radiation, there is a risk of exposure to NITI imaging in children and pregnant women.

2.5. Periapical radiographs

Intraoral periapical radiographs are frequently used to assess the depth of dental caries. CNN-based models proposed in Refs. [32–37] were developed to detect the presence of caries. Detection accuracy was 89 % for premolars, 88 % for molars, and 82 % when both were combined [32]. A back-propagation neural network [33] achieved 97.1 % accuracy, while the FRCNN model showed better precision and F1 scores than YOLOv5 [34]. A multi-input CNN ensemble model [35] achieved 99.13 % accuracy, while another method using MPCA features with a neural network reached 94.4 % [36]. A model combining variational techniques and CNN [37] detected proximal caries with an F1max of 0.74 by focusing on tooth crowns. However, all these models only detect the presence of caries, not their precise location. For segmentation, a semi-supervised model [38] attained over 50 % Mean IoU in both validation and testing.

2.6. Color oral photographs

Traditional methods for detecting dental caries utilize images such as panoramic, bitewing, CBCT, NITI, and periapical scans. These require patients to visit a radiologist, who then analyzes the images using special software, after which a dentist interprets the results. While accurate, these methods are expensive, time-consuming, and depend heavily on the skill of the professionals and the quality of the equipment.

To make caries detection faster and more affordable, researchers have started using smartphone photos of teeth. Deep learning models like DCNN [39] and Mask RCNN [40] can now detect cavities, gum disease, stains, and plaque from color dental photos. Some studies even

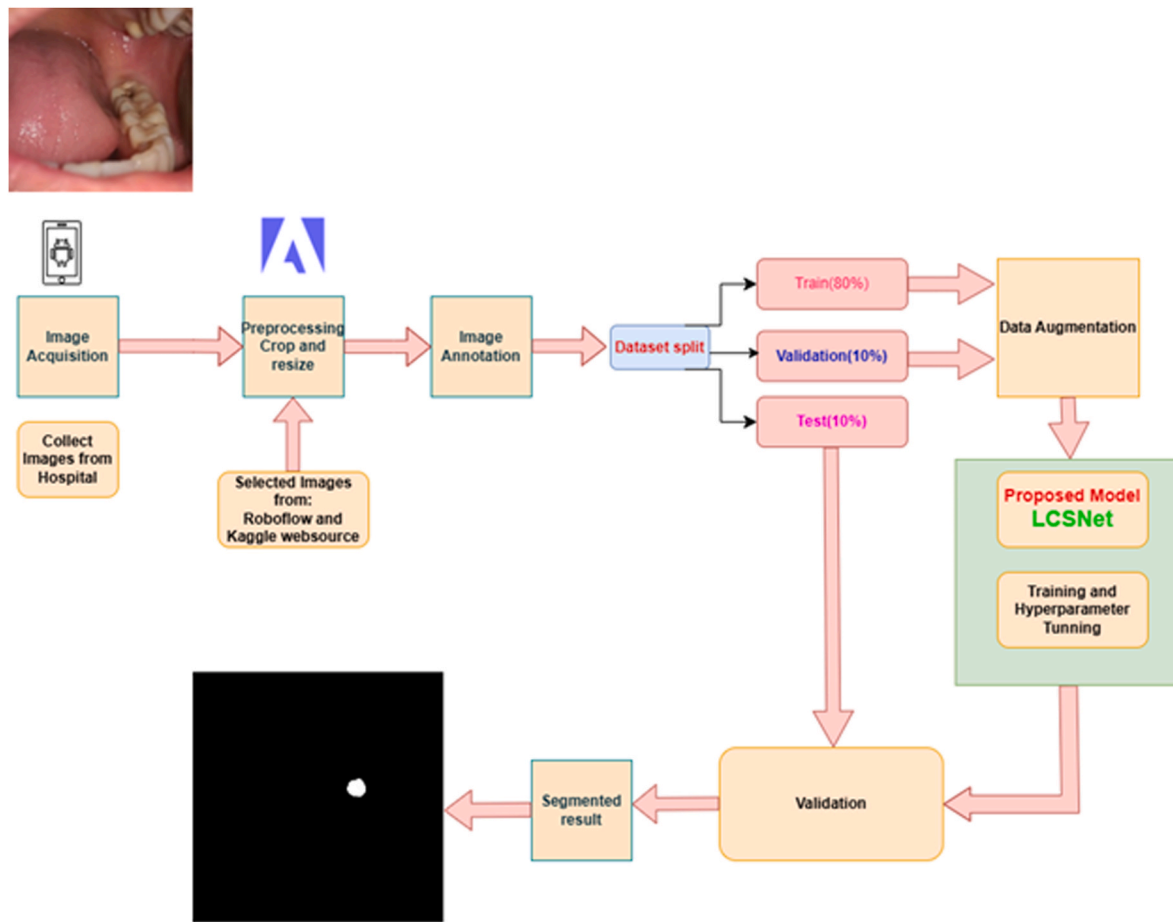


Fig. 1. Flow diagram of the proposed method.

include mobile apps that allow patients to communicate with dentists, plan treatments, and save money.

However, many of these models don't analyze deeper parts of the mouth. Some newer models use single-tooth photos to classify different types of cavities with over 90 % accuracy [41]. Other models, such as RDFNet [42] and ResNet-based models [43], improve both speed and accuracy, especially in children's molars. ConvNet [44] showed promising results, but sometimes confused stains with cavities.

Support Vector Machines [45] classify no surface change, visually non-cavitated, and cavitated dental caries on occlusal tooth surfaces by extracting single-tooth images from intraoral photographs, achieving higher specificity and sensitivity compared to four object detection models (FRCNN, YOLOv3, SSD, ResNet) proposed in Ref. [46]. Other deep learning models, such as Xception, VGG, and DenseNet, also performed well, with some achieving an accuracy of 94 % [47]. SegFormer-B5 [48], a vision transformer model, achieved very high precision in locating cavities. MRCNN [11] combined both radiographs and color photos for better detection.

Recently, two advanced methods were introduced to classify early, moderate, and severe decay using RGB images. CariesFG [62] uses detailed attention-based networks but is slower. Meanwhile, Multi-category fusion contrastive learning [63] focuses on selecting key data and separating categories more efficiently by adapting the Core Data Selection Method with Jensen-Shannon Divergence (JSCDS) [64]. JSCDS helps select the most useful data by measuring how closely samples relate to each other, improving training efficiency.

Among all these diagnostic tools, primary caries detection involves taking periapical radiographs that capture the affected teeth and some surrounding areas. Bitewing radiographs are used to visualize caries in the crowns of teeth, while panoramic radiographs help diagnose dental

conditions across the entire mouth. CBCT and NITI radiographs are rarely employed for examining dental caries. Each of these diagnostic methods requires experienced radiologists and calibrated equipment, and the radiation they emit can also affect human health. Conditions like oral hygiene and the positioning of the capturing equipment can influence the films' reflections, potentially leading to false diagnoses. To address these issues, the development of artificial intelligence (AI) has sparked research into solutions for caries detection by integrating smartphones with AI. AI-based caries detection models that use smartphone photos of dental conditions either classify teeth as caries or non-caries based on training with single extracted teeth or detect caries from external oral images, which may fail to diagnose caries in intraoral photographs.

This work proposes a pixel-level segmentation model for the early-stage detection of dental caries from smartphone camera-captured images of both outer-oral and intraoral mouth surfaces.

3. Materials and methods

In this section, the dataset and the methodology adopted in this study, as per the flow diagram given in Fig. 1, are described under subheadings.

3.1. Study protocol and dataset collection

This study was conducted at the Department of Oral Medicine and Radiology, Government Dental College and Research Institute (GDCRI) in Bengaluru. It was approved by the Institutional Ethical Committee of GDCRI (No. GDCRI/IEC-ACM(2)/12/2021-22) as well as by the Department of Oral Medicine and Radiology at DAPM R V Dental

Table 1

Distribution of the images in the dataset.

Dataset Source reference	Number of Color Images
GDCRI	344
DAPM RVDC	278

Table 2

Textural distribution of color photographs of dental Images.

Image Texture	Number of Images
Outer oral cavity	143
Upper Inner cavity	99
Lower inner cavity	186
Images with single-tooth	20
Images with two teeth	60
Images with only molar	114
Total Images	622

College in Bengaluru, which received approval from the Research Substances and Institutional Review Board Committee of DAPM RVDC (IRB No. 453/VDL-2/2022). Participants were recruited from patients visiting the hospital after being informed about the purpose and process of the study, as outlined in the approved study protocol, from December 2022 to January 2024. Oral photographs of dental caries were collected using various companies' Android phones (Motorola, Samsung, and Apple), with informed consent obtained from each patient under the supervision of an experienced dentist. The patient age range considered for the study was 12–70 years. Details regarding the number of images

collected are presented in Table 1.

3.2. Data analysis

The dataset utilized in this work comprises images of various textures, with their distribution related to different textures illustrated in Table 2. The developed model effectively segments dental caries across all textural views.

3.3. Data preparation and augmentation

All collected oral photographs are resized to 512×512 pixels using Adobe Express Resizer without disturbing the region of interest (ROI) and are stored in PNG format. The resized images are annotated using the CVAT.ai online annotation tool, an open-source application developed by Intel and licensed by MIT. The ground-truth mask is generated after annotation and contains the foreground representing the ROI (caries region) with a pixel value of 255. In contrast, the background represents the non-caries region with a pixel value of '0'. The data is split into 80 % for training, 10 % for validation, and 10 % for testing. Out of 622 collected images, 498 are training, 62 are validation, and 62 are test images. The training and validation dataset samples are increased by applying various data augmentation techniques, including rotation at angles of 90° , 180° , and 270° , random horizontal and vertical flips, Gaussian blurring, adding Gaussian noise, and histogram equalization. After data augmentation, 2988 images were used for training, and 372 images were used for validation.

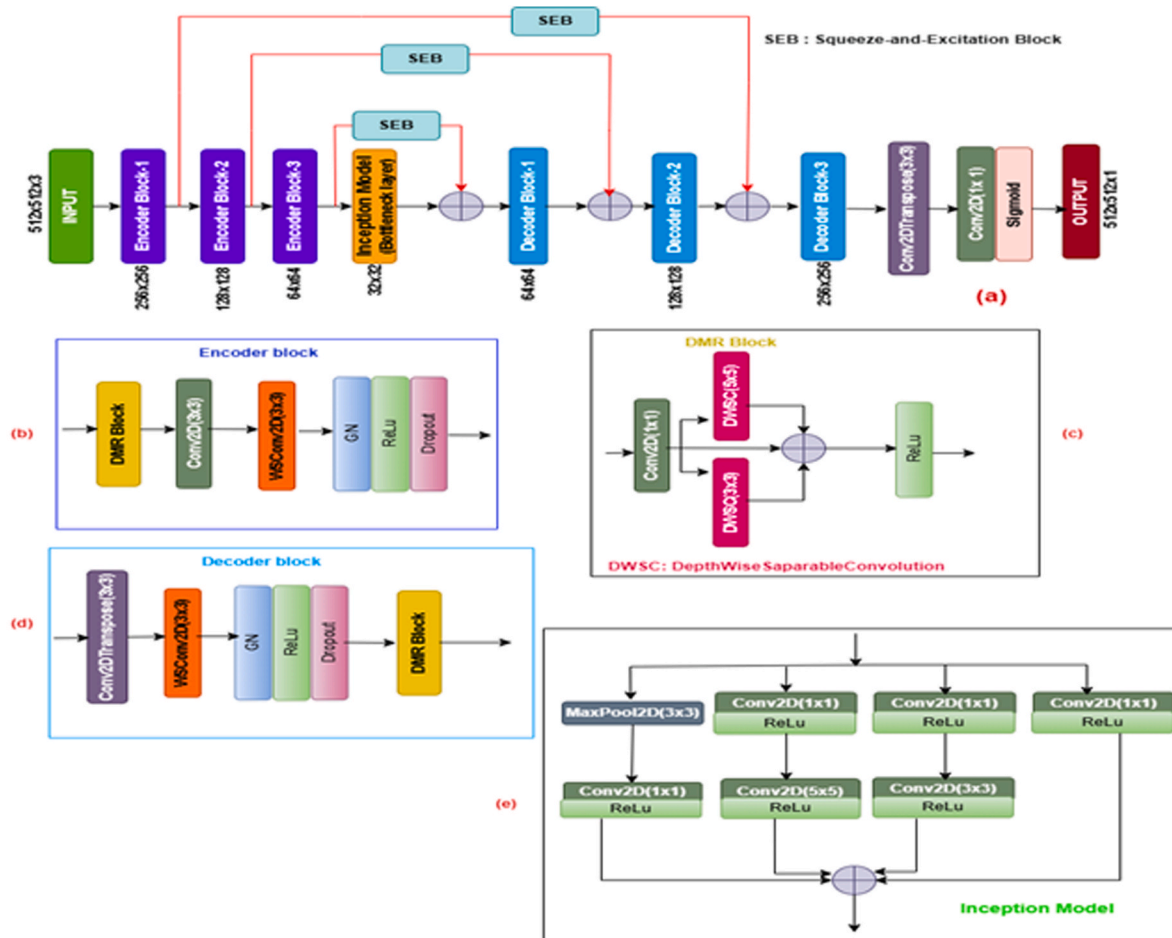


Fig. 2. Proposed LCSNet Architecture - (a). LCSNet Model (b). Encoder Block of LCSNet (c). DMR Block of Decoder (d). Decoder block of LCSNet (e). Inception model (Bottle-neck layer of LCSNet).

Table 3

Comparison of performance metrics of LCSNet for different loss functions.

Performance metrics	Loss function			
	Dice Loss + Focal Loss	Dice Loss	Dice Loss + BCE Loss	Dice Loss + Focal Loss + BCE Loss
Accuracy	97.12	97.35	97.26	97.3
Precision	70.2	73.1	74.4	71.45
Recall	65.83	70.2	63.58	69.2
F1-Score	68.4	71.14	69.2	70.6
IoU	53.2	56.8	54	55.7

3.4. Proposed LCSNet architecture

The proposed LCSNet follows the encoder-decoder architecture and consists of Dual Multiscale Residual (DMR) blocks in both the encoder and decoder, inspired by concepts used in Mini-Net [52]. The encoder block includes one DMR block followed by a 2D convolution layer, GroupNormalization with WeightStandardization [54], and a ReLU [55] activation function. The DMR block contains a 1×1 2D convolution layer, and its output is used as input for two Depthwise Separable Conv layers, one with a kernel size of 3 and the other with a kernel size of 5. The outputs from these three paths are added element-wise and passed through a ReLU activation, enabling the network to gather multi-scale information from 1×1 , 3×3 , and 5×5 receptive fields with minimal computational overhead.

The decoder block performs the reverse operation of the encoder block, using 2D transpose convolution for up-sampling the input. After the 2D transpose convolution, the decoder block includes Group-Normalization with Weight Standardization [54] and ReLU activation, concatenation with skip features, and the output from the concatenation layer is provided as input to the last layer of the decoder: the DMR block. LCSNet starts with an input layer of shape $512 \times 512 \times 3$, three encoder blocks to down-sample the data, and the pre-trained InceptionV3 model as the bottleneck layer. It features three decoder blocks followed by a 2D transpose convolution layer. The model's output is of shape $512 \times 512 \times 1$ and is obtained from the 1×1 convolution layer. With a batch size of the dataset set to 8, Group Normalization (GN) with Weight

Standardization performs well. Compared to Batch Normalization, it avoids overfitting, effectively normalizing features under small-batch constraints and standardizing convolutional kernels for consistent training.

LCSNet also employs transfer learning by utilizing the InceptionV3 [56] in the bottleneck layer, which is a pre-trained model trained on the ImageNet Large Visual Recognition Challenge (ILSVRC) 2012 dataset for image classification, enabling the model to learn feature maps more effectively, even with limited datasets. InceptionV3 comprises multiple parallel convolutional branches with varying kernel sizes, which are fused at various stages, enabling the bottleneck to capture multi-scale features efficiently. Each skip connection is enhanced by an SE block proposed in Ref. [57], which performs channel-wise attention by applying global average pooling across spatial dimensions to capture channel-wise statistics. The pooled features are processed through a reduction layer followed by an expansion layer with a sigmoid activation to generate channel-specific weights. The original feature maps are recovered by multiplying each channel by its learned weight. The SE block emphasizes the most relevant channels, enhancing the feature fusion between the encoder and decoder. Fig. 2 illustrates a comprehensive, multi-layered diagram of the proposed LCSNet. A detailed model summary of the developed LCSNet architecture is attached as a supplementary file.

3.5. Implementation details

The developed model is implemented and tested using the deep learning frameworks TensorFlow 2.15 and Keras 2.15, coded in Python 3.10.14. We used Google Colab Pro to leverage an NVIDIA A100-SXM4-40GB Graphics Processing Unit to run the model. Hyperparameters are set with a batch size of 8, 100 epochs, and an initial learning rate of $1e-4$, which decreases to $1e-8$. The learning rate is adjusted during training through the callback function ReduceLROnPlateau, which tracks validation loss. In this model, the learning rate for the backbone layers is adaptive, guided by the Learning Rate Policy (LRP) factor, set to 0.25. An LRP patience of 4 allows the scheduler to pause before updating the learning rate, preventing model overfitting. For all the models, we used

Table 4

Comparison of the performance of our proposed architecture with other deep learning models.

Deep Learning Models	Performance Metrics					Trainable parameters(in Million)
	Accuracy	Precision	Recall	F1-Score	IoU	
UNET [13]	97.59	76.82	69.2	72.6	57.8	15.334
UNET-Resnet50 [49]	97.45	78.81	60.7	66.42	52.93	17.506
UNET-Inceptionv3 [50]	97.58	82.15	62.9	69.7	56.3	14.525
UNET-InceptionResnetv2 [51]	97.69	84.3	61.1	69.8	56.9	33.615
UNET-Vgg16 [53]	97.82	82.12	68.6	75.8	62.18	25.86
TransAttUnet [66]	96.4	65.4	56.6	58.8	43.5	34.81
ResnetXt [67]	97	83.7	49.4	61.5	47.4	6.808
SelfAttention [68]	97.5	79.12	68.9	72.3	57.9	23.006
Proposed-LCSNet	97.35	73.1	70.2	71.14	56.8	5.87

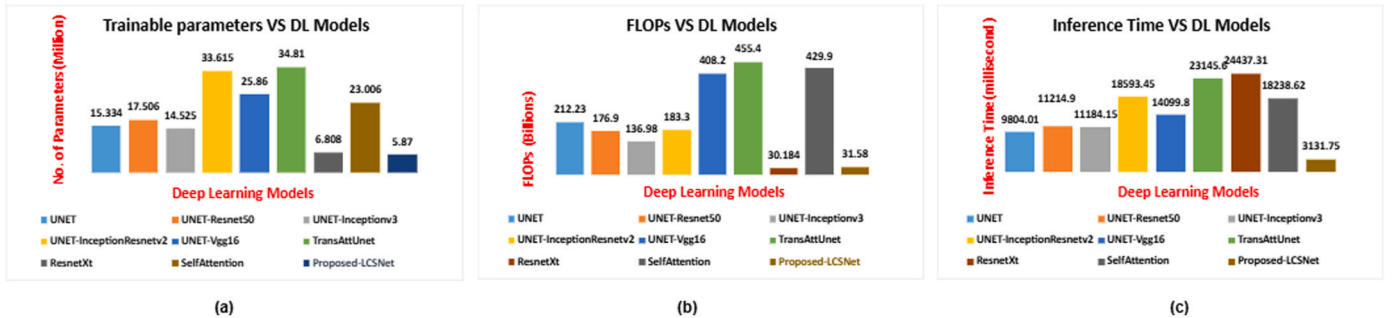


Fig. 3. Architectural complexity comparison of deep learning models: a) Based on trainable parameters b) Based on FLOPs c) Based on Inference Time.

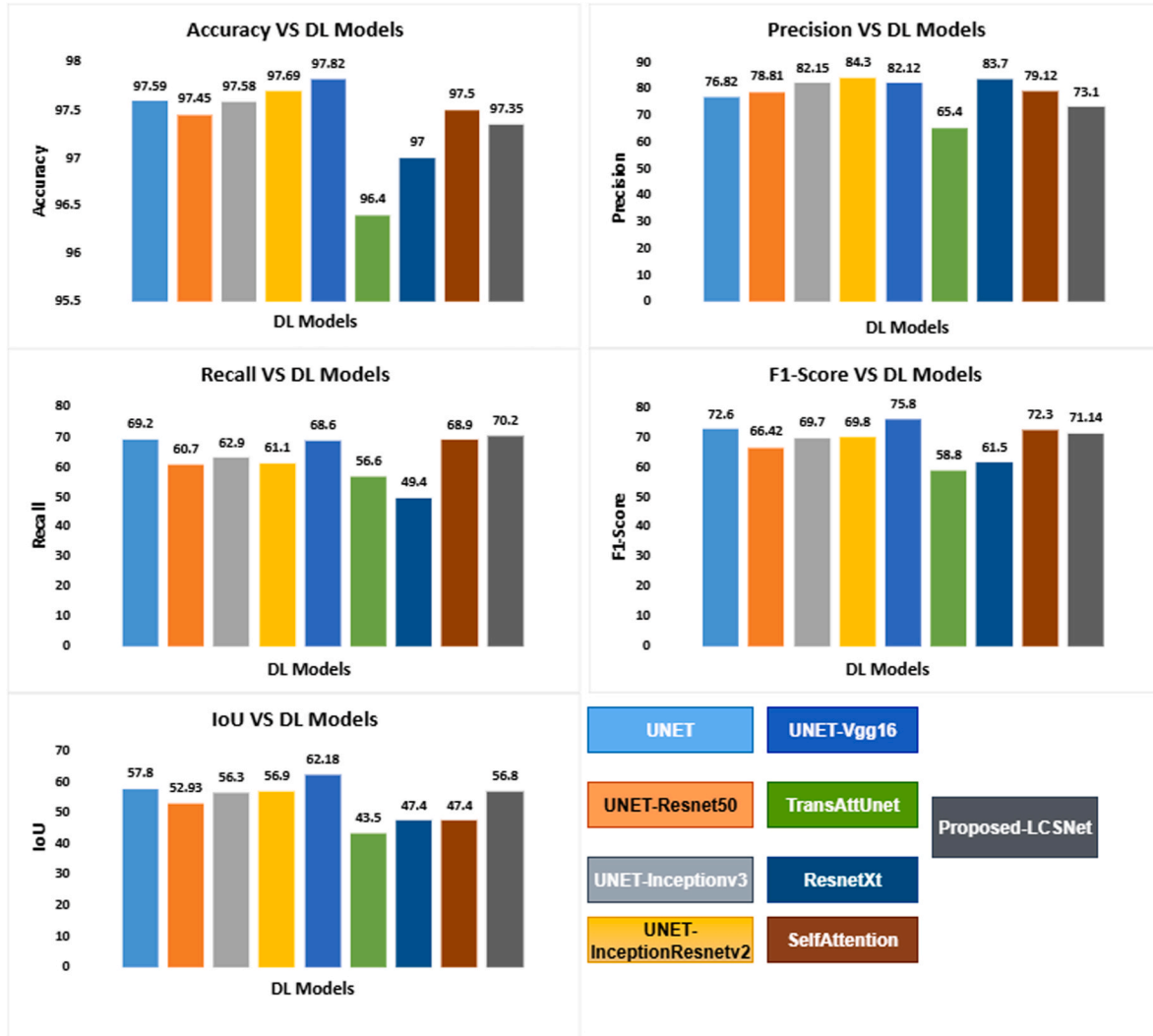


Fig. 4. Performance metrics comparison of deep learning models.

Table 5

Comparison of inference time and computational requirements of deep learning models.

Deep Learning Models	Trainable parameters(in Million)	Inference Time(in millisecond)	FLOPs(in Billion)
UNET [13]	15.334	9804.01	212.23
UNET-Resnet50 [49]	17.506	11214.99	176.9
UNET-Inceptionv3 [50]	14.525	11184.15	136.98
UNET-InceptionResnetv2 [51]	33.615	18593.45	183.3
UNET-Vgg16 [53]	25.86	14099.8	408.2
TransAttUnet [66]	34.81	23145.6	455.4
ResnetXt [67]	6.808	24437.31	30.184
SelfAttention [68]	23.006	18238.62	429.9
Proposed-LCSNet	5.87	3131.75	31.58

Adam [58], an adaptive learning rate optimizer, which also helps with parameter updating and network optimization.

3.6. Performance metrics used

The performance of the proposed model is assessed using the measured values of performance metrics: accuracy, precision, recall, F1-score, and IoU. Dice loss is used as the loss function for the developed segmentation model. These metrics are defined based on four indices: True-positive (TP), True-negative (TN), False-positive (FP), and False-

negative (FN). The mathematical relationships of these indices to each of the performance metrics are provided in equations (1)–(6), respectively.

$$Accuracy = \frac{(TP + TN)}{(TP + FP + FN + TN)} \quad (1)$$

$$Precision = \frac{TP}{TP + FN + \epsilon} \quad (2)$$

Table 6

The statistical significance of the proposed model.

Baseline Model	Mean F1-Score(Baseline Model)	Mean F1-Score(Proposed-LCSNet Model)	Mean Diff	P-Value	95 % CI	Effect Size(d)
UNET [13]	0.7919	0.8259	0.0340	0.0856	(-0.0049, 0.0729)	0.2149
UNET-Resnet50 [49]	0.7522	0.8259	0.0737	0.0013	(0.0299, 0.1175)	0.4137
UNET-Inceptionv3 [50]	0.7474	0.8259	0.786	0.0002	(0.0381, 0.1190)	0.4774
UNET-InceptionResnetv2 [51]	0.7851	0.8259	0.0408	0.0780	(0.0309, 0.0709)	0.2870
UNET-Vgg16 [53]	0.9114	0.8259	-0.0855	0.0000	(-0.1167, -0.0543)	-0.674
TransAttUnet [66]	0.6437	0.8259	0.1820	0.0000	(0.1313, 0.2327)	0.8827
ResnetXt [67]	0.8054	0.8259	0.0202	0.1232	(-0.0056, 0.0461)	0.1923
SelfAttention [68]	0.8199	0.8259	0.0058	0.6844	(-0.0224, 0.0339)	0.0503

Table 7

Accuracy of the Deep models with and without noise removal techniques.

DL Models	Accuracy of the models			
	Without Noise removal	With Noise removal techniques		
		Gaussian	Median	Bilateral
UNET [13]	95.82	95.83	95.85	95.86
UNET-Resnet50 [49]	96.37	96.19	96.66	96.45
UNET-Inceptionv3 [50]	97.77	97.54	97.60	97.48
UNET-InceptionResnetv2 [51]	97.07	97.03	97.0	96.7
UNET-Vgg16 [53]	98.60	98.56	98.57	98.52
TransAttUnet [66]	96.45	96.35	96.33	96.08
ResnetXt [67]	96.32	96.34	96.33	96.30
SelfAttention [68]	95.78	95.65	95.65	95.61
Proposed-LCSNet	97.13	97.07	97.05	97.03

model depend on the number of trainable parameters, influenced by the model's layers. In Table 4, the seventh column displays the trainable parameters of each state-of-the-art model after modifying the base architecture of these models by applying GroupNormalization with WeightStandardization and introducing a 2 % dropout in the convolution block. We implemented this modification in all the other models because the developed LCSNet utilizes GroupNormalization with WeightStandardization and a 2 % dropout rate in the convolution block to prevent overfitting. As shown in Fig. 3, the proposed deep learning model utilizes fewer trainable parameters (Fig. (a)) than all the other models, making it computationally efficient as it takes less inference time to segment dental caries in unseen data (Fig. (c)). Here, floating-point operations (FLOPs) represent the computing power of a given hardware device, like a GPU. The smaller the value (Fig. 3(b)), the faster the computing speed.

4.2. Evaluation metrics

We tested all the other models with the dataset used in this study. Quantitative analysis was performed on the performance metrics obtained from the validation dataset. As shown in Table 4, the recall value of LCSNet is more significant than that of the other segmentation models and was trained with only 5.87 million parameters; the IoU value achieved is 56.8 %, which surpasses that of UNET-InceptionV3, UNET-Resnet50, TransAttUnet, and ResnetXt models, each trained with 14.525, 17.506, 34.81, and 6.808 million parameters, respectively. The accuracy of LCSNet (97.35 %) is comparable to these models. The measured recall value of the developed LCSNet is 70.2 %, which exceeds the recall values of other models. The F1-Score measured was 71.14 %, higher than that of UNET-Resnet50, UNET-InceptionV3, UNET-InceptionResnetV2, TransAttUnet, and ResnetXt. The precision value measured is lower than all the other models except TransAttUnet. Since F1-Score and IoU are key performance metrics in segmentation tasks for comparing models, the proposed model outperforms all others. Although models like UNET [13], UNET-Vgg16 [53], and SelfAttention [68] achieved higher F1-Score and IoU values, their performance was slower and less efficient in terms of computational resources. The performance metrics comparison for each deep learning model is illustrated using a bar graph, as shown in Fig. 4.

As shown in Table 5, the proposed LCSNet uses fewer trainable parameters than other deep learning segmentation methods. This is because LCSNet is built using Inception and SE blocks, along with Group Normalization and Weight Standardization. The proposed LCSNet model takes 3131.75 ms for inference and requires only 31.58 billion FLOPs. These results indicate that the proposed LCSNet model is both faster and more efficient in terms of computational resources.

The statistical significance of the proposed model is shown by considering the Mean F1-score across all models, as presented in Table 6. The test dataset was used to measure the Mean F1-Score. The proposed model (LCSNet) consistently outperforms most baseline models with statistically significant improvements, especially over models InceptionV3, ResNet50, and TransAttUnet, where p-values are well below 0.05 and effect sizes are moderate to large. Model Vgg16 is the only one that significantly outperforms the proposed model, with a higher mean

$$Recall = \frac{TP}{TP + FP + \epsilon} \quad (3)$$

$$F_1 - Score = 2 \frac{Precision \cdot Recall}{Precision + Recall + \epsilon} \quad (4)$$

$$IoU = \frac{\sum (y_t * y_p)}{\sum y_t + \sum y_p - \sum (y_t * y_p) + \epsilon} \quad (5)$$

$$Dice Loss = 1 - \frac{2 * \sum (y_t * y_p) + smooth}{\sum y_t + \sum y_p + smooth} \quad (6)$$

Where: y_t - represents the Ground truth mask

y_p - represents the predicted mask

ϵ - small positive constant used for numerical stability taken as 1×10^{-7}

smooth - A small constant is added to prevent division by zero.

The developed LCSNet model also experimented with various combinations of loss functions, including Dice loss + Focal loss, Dice loss + Focal loss + binary cross-entropy (BCE), Dice loss + BCE loss, and only Dice loss. As shown in Table 3, the measured performance metrics are higher for the Dice loss than for the other loss functions. However, the precision value is greater when the model is trained with Dice loss + BCE loss.

4. Results and discussion

This section presents a performance comparison of LCSNet with other state-of-the-art models: UNET, UNET-VGG16, UNET-Resnet50, UNET-InceptionV3, UNET-InceptionResnetv2, TransAttUnet, ResNetXt, and SelfAttention in terms of architectural complexity, evaluation metrics, and prediction results.

4.1. Architectural comparison

The memory requirements and execution time of any deep learning

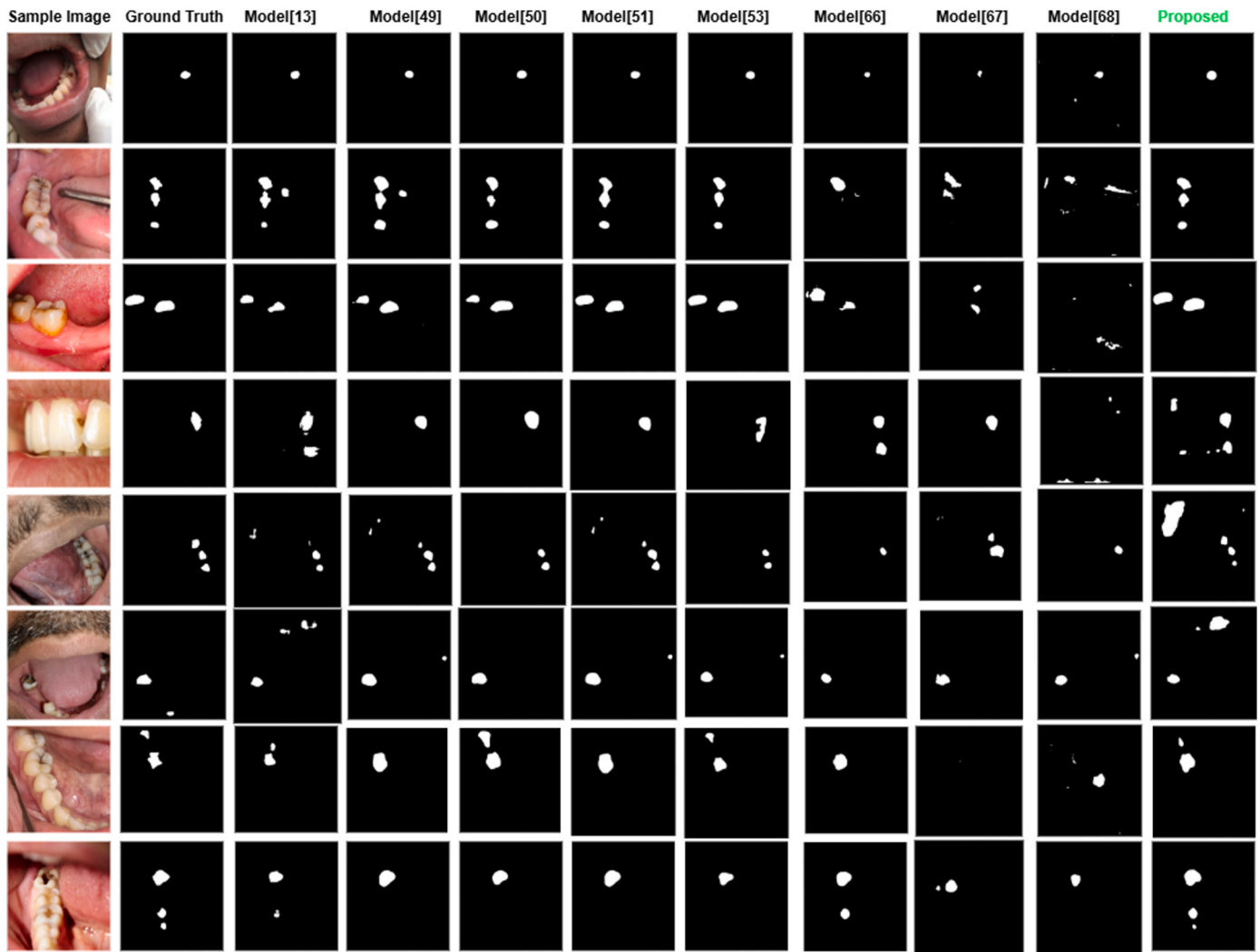


Fig. 5. Segmentation results of each model: The Ist row represents the Sample Images, the IInd row presents the ground truth, IIIrd to Xth rows represent the prediction results of state-of-the-art models, and the last row presents the predicted results of LCSNet(proposed).

F1-Score and a negative effect size. Overall, the proposed model demonstrates strong statistical significance and practical relevance compared to most alternatives.

We have also performed an accuracy calculation on the test dataset with and without applying noise removal techniques, and the results are presented in Table 7. The noise removal techniques adapted in this work to compare the achieved accuracy of the proposed model with the other models are: Gaussian, median, and bilateral preprocessing.

Across all models, the accuracy values stay very close, with differences ranging from 0.01 to 0.3 %. This suggests the models are resilient to noise. The proposed model maintains a strong accuracy of around 97 %, showcasing its robustness and competitiveness compared to other models.

4.3. Prediction results

The prediction results of each model were analyzed qualitatively for eight sample images representing different dental caries conditions. As presented in Fig. 5, the first row represents the sample images, the corresponding ground-truth mask is shown in the second row, and subsequent rows represent the corresponding prediction results of each model. As depicted in Fig. 5, for most of the sample images, the segmentation results of the proposed LCSNet match the ground truth. Although the model is trained on ground truth images with blunt edges,

the predicted segments exhibit sharp edges. Comparatively, the prediction results of SelfAttention [68] do not match the ground truth; it also detected non-carries regions. Integrating an SE block in the skip connection and InceptionV3 at the bottleneck layer of the LCSNet model enables it to effectively learn latent features from the blunt and unclear edges found in low-illumination regions representing the deep inner-oral cavity, as shown in Fig. 5.

4.4. Limitations

As presented in Table 2, the dataset used in this study contains very few samples of single teeth, two teeth, and upper inner oral cavity images. The developed model was not trained with sufficient samples, leading to a reduced IoU for the LCSNet. Training the model with more samples of single teeth, two teeth, upper inner oral cavity images, and additional images representing unclear edges could enhance the IoU and other performance metrics. Overserving the predicted result, the model also predicts other black regions of the images as caries regions in some samples. To overcome this, the model needs to be trained with more images of this kind. Due to the absence of a public dataset, our study focuses on a specific dataset we prepared for this research, limiting our findings' applicability. In future research, we will evaluate the performance of the developed LCSNet model on various datasets collected from different sources. This will enable us to verify the model's

Table 8
Checklist comprise of planning, conducting and reporting of AI studies in dental research.

Item No.	Planning and conducting	Agreement	Item No.	Planning and conducting	Agreement
1	Study Goal	9	14b	Data protection	8
2	Study Focus	10	14c	Missing Data	6
3	Data	6	14d	Data preprocessing	9
4	Study Aim	8	15	Reference Test	7
5	Reference Test	9	16	Sample Size	6
6	Clustering	6	17	Model	10
7	Test Data	10	17a	Model parameters	10
8	Computational Resources	10	18	Training	10
9	Comparators	10	19	Justify the best- performing model	10
	Reporting		20	Evaluation	10
10	Title	10	21	Uncertainty	8
11	Abstract	7	22	Explainability	6
12	Introduction	10	23	Results	10
13	Study design	10	23a	Performance Metrics and Data Partitions	8
14	Data	8	24	Discussion	10
14a	Sampling	7	25	Other	10

generalization and reliability.

The pros and cons of our study were assessed by mapping them to the measurable checklist provided in paper [65], as shown in Table 8. The evaluation indicates that the proposed research scores highly in key areas such as study design, data reporting, model development, training, evaluation, and discussion, aligning closely with established guidelines for high-quality AI research in dentistry. Although some aspects, like data clustering, reference test construction, and reporting of uncertainty and explainability, could be improved, these do not compromise the overall robustness of the work.

5. Conclusion

Automatic segmentation of dental caries aids in detecting and localizing dental caries within images. The imaging modality used in this study helps to avoid radiographs, thereby reducing treatment costs and the likelihood of false diagnoses. In this paper, we present a new lightweight deep-learning model for dental caries segmentation using smartphone camera photographs of dental conditions. The proposed lightweight deep-learning model, LCSNet, addresses architectural complexity, lower execution times, overlapping dental caries regions, and masks with blunt edges. The Group Normalization with Weight Standardization in the convolution block prevents overfitting even with lower batch sizes. The InceptionV3 bottleneck layer effectively captures multi-scale features. Depth-wise separable convolutions in the DMR block reduce the parameter count and computational costs, enhancing the architecture’s efficiency. The SE block applies global average pooling across spatial dimensions to capture channel-wise statistics. With all these features, the proposed novel lightweight model segments dental caries in smartphone camera photographs of oral conditions, even in low-resolution images. Experimental results are compared with those of other deep learning models. The proposed model requires less inference time to segment caries on unseen datasets and also requires fewer computational resources due to reduced floating-point operations (FLOPs). However, a public dataset is not available to test the performance of the proposed model. The performance of the proposed model could be improved by increasing the dataset and adapting different models in the bottleneck layer and the skip connection.

CRediT authorship contribution statement

R.C. Radha: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Data curation, Conceptualization. **B.S. Raghavendra:** Writing – review & editing, Visualization, Supervision, Methodology, Formal analysis, Conceptualization. **Rishabh Kumar Hota:** Software, Resources, Methodology,

Conceptualization. **K.R. Vijayalakshmi:** Validation, Supervision, Data curation, Conceptualization. **Seema Patil:** Validation, Supervision, Data curation, Conceptualization. **A.V. Narasimhadhan:** Writing – review & editing, Visualization, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

“The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper”.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ibmed.2025.100304>.

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